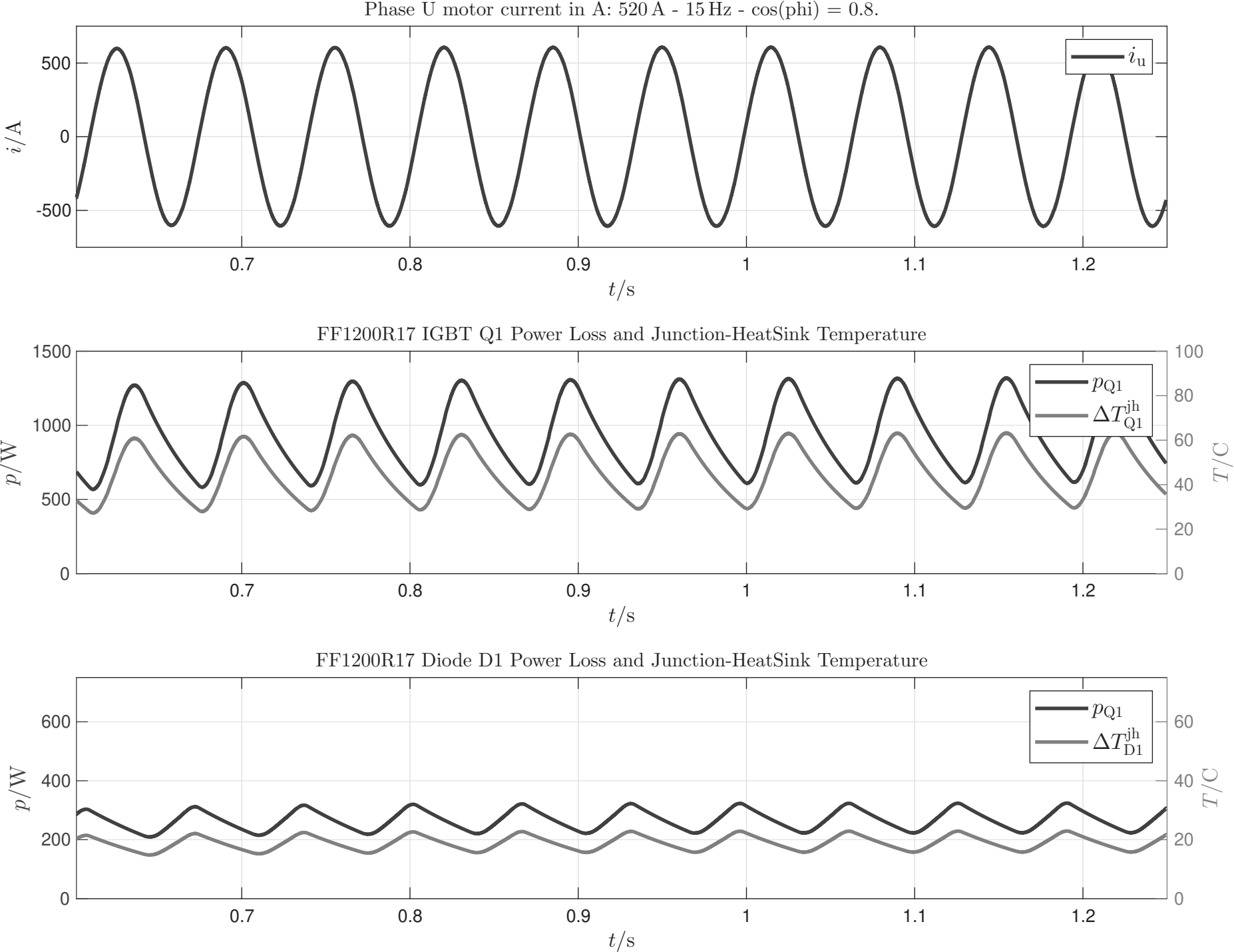

Phase U motor current in A: 520 A - 15 Hz - cos(phi) = 0.8.
i_u
i/A
500
0
-500
0.7
0.8
0.9
1
1.1
1.2
t/s
FF1200R17 IGBT Q1 Power Loss and Junction-HeatSink Temperature
p_{Q1}
ΔT^{jh}_{Q1}
p/W
T/C
1500
1000
500
0
100
80
60
40
20
0
0.7
0.8
0.9
1
1.1
1.2
t/s
FF1200R17 Diode D1 Power Loss and Junction-HeatSink Temperature
p_{Q1}
ΔT^{jh}_{D1}
p/W
T/C
600
400
200
0
60
40
20
0
0.7
0.8
0.9
1
1.1
1.2
t/s